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Original Research Article

Detection of vancomycin variable enterococci (VVE) among clinical isolates of *Enterococcus faecium* collected across India-first report from the subcontinent

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ABSTRACT

Purpose: Emergence of vancomycin variable enterococci (VVE) poses a challenge to empiric vancomycin therapy. Vancomycin-variable enterococci (VVE) are *van*A-positive, yet phenotypically vancomycin-susceptible enterococci that can switch to a vancomycin-resistant phenotype when exposed to vancomycin. The aim of the present study was to determine the prevalence of VVE in India.
Methods: Isolates of phenotypically vancomycin susceptible *Enterococcus faecium* from 20 tertiary care hospitals across India were collected and tested for the presence of *van*A, *van*R, *van*S, *van*B and *van*C genes by conventional PCR using previously published primers. Isolates positive for *van*A gene were considered as VVE.
Results: The prevalence of VVE was 1.5% (5/340). Only one VVE isolate was positive for *van*R and *van*S, and all the isolates were negative for *van*B and *van*C.
Conclusions: Although the prevalence is low, our finding emphasizes the importance of routinely screening for *van* genes in enterococci that are phenotypically susceptible. Silenced *van*A able to escape detection and revert to resistance during vancomycin therapy represents a new challenge in clinical settings.

1. Introduction

Enterococci are opportunistic pathogens that are a major cause of hospital-acquired infections such as septicaemia, urinary tract infections, endocarditis, and blood stream infections [1–3]. Of the many species of enterococci, *E.faecium* is the most resistant as well as virulent. Increased pathogenicity in this species is due to an extension of hospital-adapted genetic lineages with more tolerance and virulence traits than commensal enterococci. Mobile elements are often used to encode certain traits [4,5].

The *van*A, *van*B, *van*D, *van*E, *van*G, *van*L, *van*M, and *van*N are eight gene clusters linked to acquired vancomycin resistance in enterococci [6,7]. The most common resistance mechanism, mediated by *van*A confers high-level resistance by substituting the glycopeptide binding site in the peptidoglycan precursor termini from D-alanine to D-lactate by *van*H, *van*A, and *van*X activities [8,9]. During glycopeptide exposure, *van*S regulates this mechanism by phosphorylation and subsequent attachment of the *van*R activator to particular upstream regions of the *van*RS and *van*HAX promoters [10].

Recently, strains of enterococci carrying the *van*A gene but susceptible to the drug by phenotypic tests were identified in Canada, South Korea, Denmark and Norway [11]. These isolates labelled as vancomycin variable enterococci (VVE) may pose grave clinical implications because of their ability to revert to a fully resistant phenotype on exposure to vancomycin. Moreover, these isolates do not grow on media designed to select vancomycin resistant enterococci thereby escaping detection.

VVE identification is therefore critical in clinical samples to ensure appropriate antibiotic theraphy and in screening samples to prevent nosocomial spread. There have been no reports of VVE from India to date, and hence this study was undertaken to detect vancomycin variable enterococci (VVE) among clinically significant isolates.

2. Materials and methods

The study was conducted in the department of Microbiology after approval by the Institutional Ethics Committee for Observational Studies.

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During the study period of 18 months, from March 2019 to August 2020, all consecutive, non-repetitive isolates of *E. faecium* from clinical samples sent to our laboratory were included. Our centre is a nodal centre under Indian Council of Medical Research (ICMR) for surveillance of antimicrobial resistance in staphylococci and enterococci (ICMR AMRSN network) and receives isolates of enterococci from 19 other tertiary care hospitals across India. *E. faecium* isolates from these centres received during the study period were also included. The isolates were speciated using Matrix-assisted laser desorption ionization-time of flight (MALDI-TOF) mass spectrometry (MS) platforms (Vitek MS, bioMérieux, France). The demographic details were collected from investigation forms (JIPMER) or ICMR AMRSN data portal (for other centres). Antimicrobial susceptibility test was performed against ampicillin (10 μg), high-level gentamicin (120 μg), linezolid (30 μg), teicoplanin (30 μg), and vancomycin (30 μg) by Kirby-Bauer disc diffusion method (Bio-Rad, Germany) and interpreted according to CLSI 2019 guidelines [12]. The MIC of vancomycin was determined only for the VVE isolates by Etest (bioMérieux, France) and interpreted as per CLSI 2019 guidelines. All vancomycin susceptible isolates were tested for the presence of *van*A and *van*B genes by PCR [13]. DNA was extracted using the Mericon Bacteria Plus DNA extraction kit (Qiagen, Hilden, Germany). Among the isolates positive for *van*A gene, uniplex PCR was used to detect *van*R and *van*S, by employing previously published primers and protocols [14]. *E.faecalis* ATCC 51299 was used as a positive control in the PCR reactions.

3. Results

During the study period a total of 340 isolates of vancomycin susceptible *E.faecium* were included in the study (130 from our center and 210 from other centres). Of these, five isolates were positive for the *van*A gene, thereby the prevalence of VVE was 1.5%. All the isolates were negative for *van*B and *van*C while only one VVE isolate was positive for *van*R and *van*S (Figs. 1 and 2). Ampicillin and high level gentamicin resistance was common among these VVE isolates (4/5 were resistant) (Table 1).Among these five isolates, three were from Mumbai (western India), one each from Puducherry (south India) and Chandigarh (North India). The clinical and demographic details of the patients are depicted in Table 2. MIC of vancomycin for these 5 isolates ranged from 0.5 μg/ml (one isolate) to 1 μg/ml (4 isolates).

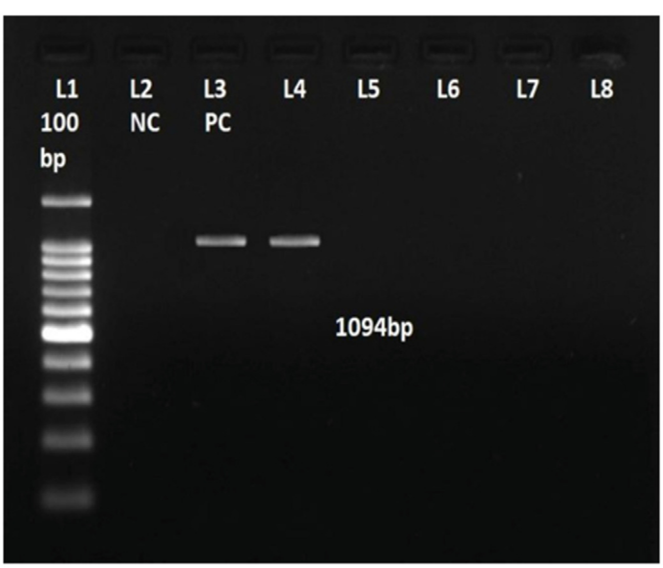

Fig. 1. Gel image for *van*S PCR (1094bp).
L1: 100bp Ladder, L2: Negative control, L3: Positive control (VRE). L4: Test VVE strain positive for *van*S, L5, 6, 7, 8: Test VVE strains (all negative for *van*S).

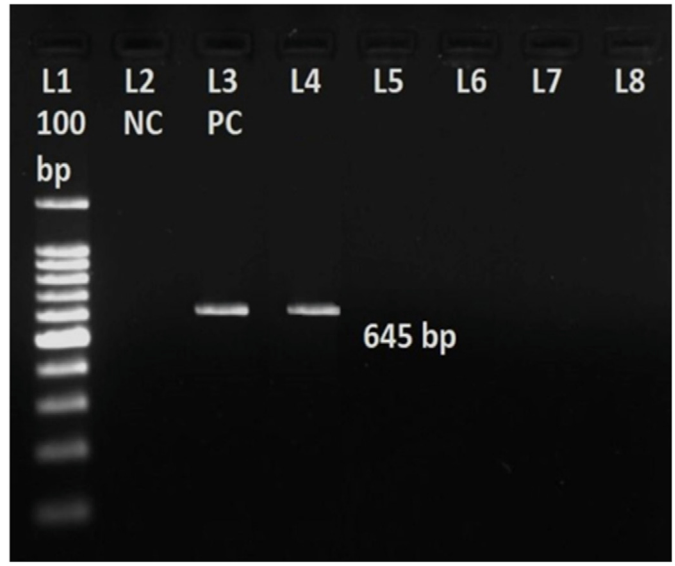

Fig. 2. Gel image for *van*R PCR (645bp)
L1: 100bp Ladder, L2: Negative control, L3: Positive control (VRE). L4: Test VVE strain positive for *van*R, L5, 6, 7, 8: Test VVE strains (all negative for *van*R).

Table 1
Antimicrobial susceptibility patterns for VSE and VVE.

Isolates	Antibiotic sensitivity (by disc diffusion)			
	Ampicillin (10 μg)	High level gentamicin (120 μg)	Teicoplanin (30 μg)	Linezolid (30 μg)
VSE (335)	46 (13.7%)	187 (55.8%)	335 (100%)	334 (99.7%)
VVE (5)	1	1	5	5

4. Discussion

VVE isolates were first identified in Canada as early as 2011. Six isolates were found to be *van*A positive and phenotypically susceptible to vancomycin. These isolates did not harbour the regulatory elements and hence it was assumed that the absence of *van*RS led to the susceptible phenotype [15]. In 2013, Jung *et al.,* found 18 (7.4%) VVE isolates out of 244 *E. faecium* isolates from faecal specimens collected from various tertiary care hospitals in Korea. Interestingly all these 18 isolates harboured the *van*RS genes and 22.2% converted to VRE when exposed to the drug [16]. In our study, however, only one isolate had harboured both these genes.

The prevalence of VVE among *van*A positive *E. faecium* was found to be 47% in a recent study from Ontario, Canada. Their significance stems from their propensity to return to a vancomycin-resistant phenotype when exposed to the antibiotic, resulting in treatment failure [17]. Furthermore, VRE detection methods almost always rely on phenotypic methods to screen out phenotypically resistant enterococci, leaving VVE

Table 2
Clinical and demographic details of the VVE patients (n = 5).

Sl.No	Centre (No. of isolates tested)	Age/ Sex	Diagnosis	Sample
1	Puducherry (1/130)	6 m/F	Chronic otitis media	Ear swab
2	Chandigarh (1/10)	9 m/M	PUO	Blood
3	Mumbai (3/24)	67 y/F	Multiple myeloma, COPD	Blood
4		37 y/F	SLE with sickle trait	Blood
5		42 y/M	Obstructive jaundice	Bile

isolates unidentified. This has serious implications for infection control, particularly in ICU setting.The *van*A operon, which includes *van*A as well as genes involved in its expression and regulation, is frequently linked to transposons like Tn1546 [15]. The regulatory apparatus is encoded by the two-component device *van*R (response regulator) and *van*S (sensor kinase). Despite having the HAX gene cassette, 7 VVE have been found to lack the *van*R and *van*S genes. These genes may exist in a few isolates, but they may be silenced.

Multiple mechanisms have been implicated in VVE/VRE in enterococci. Comparative whole-genome sequencing found that VVE strains could revert to a resistant phenotype through the *van*RS-independent constitutive expression of the resistance gene cassette. This may happen through a series of changes taking place in the region upstream of the resistance genes (like deletion of a likely transcription inhibitory secondary structure) or introduction of a new unregulated promoter (may occur during antibiotic therapy), which may lead to treatment failure. Similarly, the reversion to susceptible phenotype is observed only when small deletions occur in *van*R/S genes or these genes being silenced via IS elements (ISL3 and IS1542) as identified in a few studies [18–20].

Studies on VVE with *van*B type resistance are extremely rare. A Japanese study conducted in 2018 characterized 19 isolates of *E. faecium*, which were phenotypically vancomycin susceptible but *van*B-positive. *van*B type glycopeptide resistance gene confers resistance to vancomycin alone over a wide range of MIC levels (MIC = 4–1000 mg/L). Notably, due to mutations in or outside the *van*B gene cluster, these isolates have reverted to a resistant phenotype when exposed to vancomycin. Hashimoto *et al.*, called these strains "stealthy" *van*B-type *E. faecium* strains [21]. In the present study, however, none of the isolates were positive for *van*B.

There are reports indicating outbreaks due to this less studied variant. The first outbreak of VVE was reported in 2013 by Szakacs *et al.*, where 75% of the VVE isolates tested for clonality were identical and the remaining were highly similar. In another report in 2016, four of the VVE isolates were clonally associated and led to treatment failure [22].

VVE is mostly studied on *E. faecium*, but there have been very few reports of vancomycin variable *E. faecalis* as well. Jung *et al.*, also studied the possibility of genetic transfer of the mutant *van*A cluster across different species of enterococci. Another study conducted in Norway by Sivertsen *et al.,* included one VVE *E. faecalis* isolate to investigate the *in-vivo* horizontal transfer of a mobile genetic element containing this particular *van*A variant. They found similar IS-element insertions in its *van*A gene cluster of this isolate confirming their above hypothesis [23]. Also, in a VVE surveillance study conducted in Denmark, 1.3% of isolates belonged to *E. faecalis.* The lack of a more accurate microbiologic comparison of VVE isolates (plasmid sequencing, whole genome sequencing, etc.) to record similarities and discrepancies between strains is a limitation. VVE may also occur in strains of different genetic background, and by a variety of different molecular mechanisms, such that both virulence and the risk of selection for vancomycin resistance among VVE strains may differ [24].

The ability of VVE to cause outbreaks and treatment failure apart from disseminating antimicrobial resistance is an important aspect while conducting antimicrobial resistance surveillance in enterococci. Further characterisation by whole genome sequencing is planned in the project and may help us understand the mechanisms pertaining to VVE in the country.

5. Conclusion

To conclude, this is the first report of VVE in India, with a total prevalence of 1.5%. This alarming/ worrisome finding highlights the importance of screening for *van* genes in *Enterococcus faecium* isolates that are phenotypically susceptible as part of routine surveillance although it may not be cost effective to include this test on a routine basis.

Ethics approval

The study was approved by Institutional Ethics Committee for Observational Studies (JIP/IEC/2019/393)

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CRediT authorship contribution statement

Lakshmi Shree Viswanath: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Visualization, Writing – original draft. **Madhan Sugumar:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Sreeram Chandra Murthy Peela:** Formal analysis, Investigation, Methodology, Validation, Writing – review & editing. **Kamini Walia:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Sujatha Sistla:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

None.

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